

People and the Biosphere

Abiotic = The non-living parts of an ecosystem.

Biotic = The living parts of an ecosystem.

Boserup - theory = A theory that as the population size approaches the point when food and resources may run out, then human ingenuity will find a way of increasing production.

Malthus - theory = A theory that human population would grow faster than food or resources supply, and a disaster would then take place.

Biome = A world-scale ecosystem e.g. the tropical rainforest.

Resources (natural) = Materials that are found in the environment that are used by humans, including land, water, fossil fuels, rocks and minerals and biological resources like timber and fish.

Hydrological cycle = The movement of water between its different forms; gas (water vapour), liquid and solid forms. It is also known as the water cycle.

Indigenous people = The original human inhabitants of an area such as the rainforest that still rely on the biosphere goods such as food and medicine.

Forests under Threat

Acid precipitation = When industrial air pollution causes water vapour in the atmosphere to become acidic and fall as acid precipitation.

Ecosystem stress = Factors that put parts of the biosphere under pressure such as climate change affecting the rainforest and plant species, not able to adapt to hotter and drier conditions.

Ecotourism = Tourism that helps local communities and minimises damage to the environment.

Global actions = International organisations that try to create agreements to protect aspects of the biosphere such as CITES and REDD.

Sustainable (forest) management = Aims to prevent damage to forests in a way that helps benefit local people.

Sustainable forestry = Able to continue without causing damage to the environment.

Nutrient cycling = The transfer of nutrients around different parts of an ecosystem.

Productivity (of ecosystems) = The productivity of an ecosystem is a measure of its biomass (all its biotic components). These differences are largely due to climate.

Consuming Energy Resources

Affluence = Greater wealth or abundance.

Conventional oil & gas = Where gas & oil are extracted from accessible gas & oil fields.

Unconventional oil & gas = Where gas and oil are less accessible and requires an alternative approach to extraction such as fracking.

Biofuels = Fuel made from plant oils to power diesel vehicles.

Carbon footprints = Measurement of all the greenhouse gases as individual produces expressed as tonnes (or kg) of carbon dioxide equivalent.

Ecologically sensitive areas = Fragile ecosystems that are at risk from contamination from gas and oil extraction.

Fracking = Drilling into Earth using high-pressure water to release gas trapped in rocks.

Landscape scarring = Damage to the landscape caused by human activity such as mining,

Energy security = Access to affordable and reliable sources of energy. Countries such as Russia and Canada, with surplus energy, are energy secure.

Environmental groups = NGOs like Greenpeace that have strong views on energy futures, arguing for a more sustainable approach to energy consumption.

Non-renewable = Energy that cannot reproduce such as coal, oil and natural gas.

Renewable = A natural source of power that will never run out.

Recyclable energy = Energy resources, including biofuels and nuclear, that can be reused, so will last into the future.